Bellabeat Wellness App

New Features Suggestions based on Smart Device Usage Patterns

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Goal for our discussion today

1

Purpose statement and Dataset overview

We will share our objective and briefly talk about the dataset

2

Examine smart device usage patterns

Understanding how people using smart devices can help enhancing Bellabeat Wellness App

3

Conclusion and Recommendations

We will share our recommendations based on our findings

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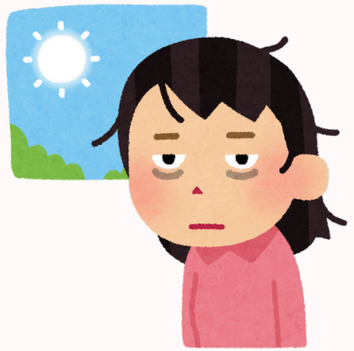
Problem



People are not getting enough sleep

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How much sleep is enough?

- Expert recommend that adults sleep between 7 and 9 hours a night
- 44% of our participants are sleep deprived

What happen if you don’t get enough sleep?

- It's estimated that drowsy driving is a factor in about 100,000 car accidents each year, resulting in about 1,500 deaths.
- 48% increased risk of heart disease and 3x higher risk of death in people with heart disease
- 83% greater risk of heart-related death and 2x higher risk of death in people with high blood pressure or diabetes
- 2.5x higher risk of developing Type 2 Diabetes in people sleeping less than 5 hours
- 2.5x more likely to have frequent mental distress with less than 6 hours of sleep per night
- Up to 40% reduction in cognitive performance with 5 hours of sleep or less

Dataset Overview

Dataset: FitBit Fitness Tracker Data

Source: Kaggle

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Period: 04/12/2016 - 05/12/2016

From: Consented 30 eligible Fitbit users

Contains: minute-level output for physical activity, heart rate, sleep monitoring and steps

Disclaimer: data does not pass the ROCCC process; however, still provides good insights

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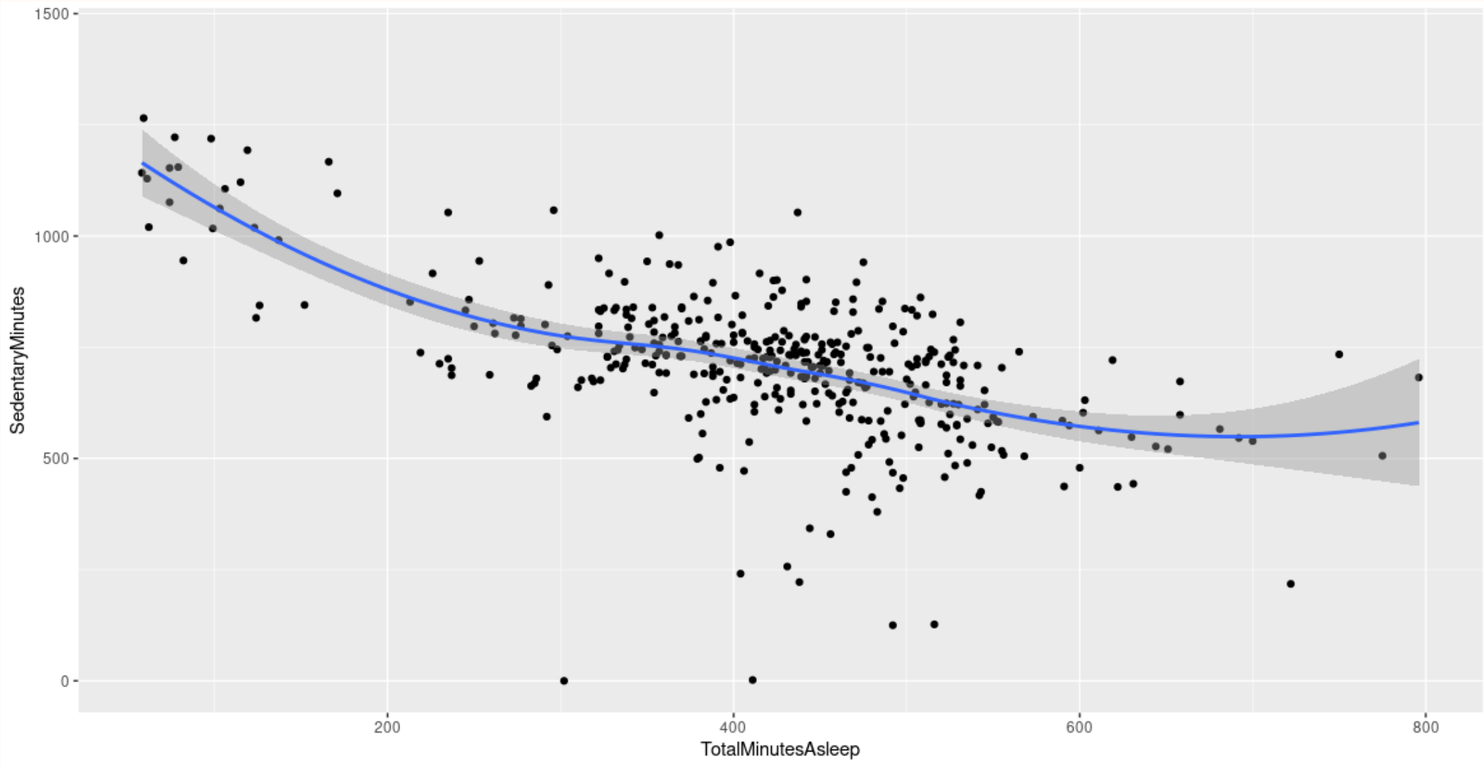
- Majority participants have 5 to 9 hours (300 - 540 minutes) of sleep per night

- Negative correlation between sedentary minutes and minutes asleep

- Among people with more than 16 sedentary hours (960 minutes) a day, there are 75% of them have less than 3 hours (180 minutes) of sleep every night

- More active group tends to have better sleep quality



→ Less daily activity leads to hard time asleep at night

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- Average steps per day is below recommended according to **National Institutes of Health**
- 8k steps a day has 50% lower risk of dying from any cause
- 12k steps a day has 65% lowers risk of dying from any cause
- Average sedentary minutes is extremely high (16.5 hours), approximately ⅔ of the day
- Averagely around 7 hours (420 minutes) of sleep per night

TotalSteps	TotalDistance	SedentaryM	VeryActiveM	FairlyActiveM	LightlyActiveM
Min. : 0	Min. : 0.000	Min. : 0.0	Min. : 0.00	Min. : 0.00	Min. : 0.0
Median : 7406	Median : 5.245	Median :1057.5	Median : 4.00	Median : 6.00	Median :199.0
Mean : 7638	Mean : 5.490	Mean : 991.2	Mean : 21.16	Mean : 13.56	Mean :192.8
Max. :36019	Max. :28.030	Max. :1440.0	Max. :210.00	Max. :143.00	Max. :518.0

TotalSleepRecords	TotalMinutesAsleep	TotalTimeInBed
Min. :1.00	Min. : 58.0	Min. : 61.0
Median :1.00	Median :432.5	Median :463.0
Mean :1.12	Mean :419.2	Mean :458.5
Max. :3.00	Max. :796.0	Max. :961.0

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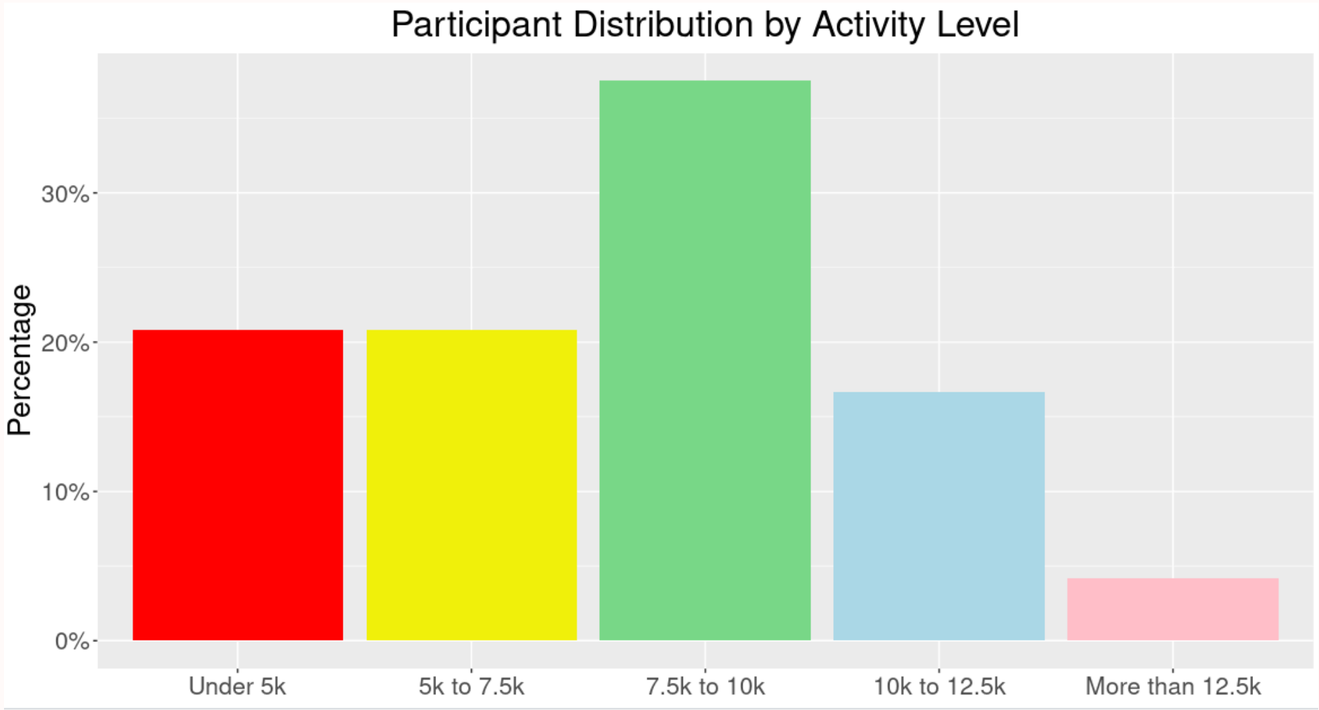
- Around 40% participants are taking less than 7.5k steps a day, less than the NIH recommendation.
- More than 35% are taking 7.5k to 10k steps a day
- Under 25% are taking more than 10k steps a day



→ Majority of participants are fairly active

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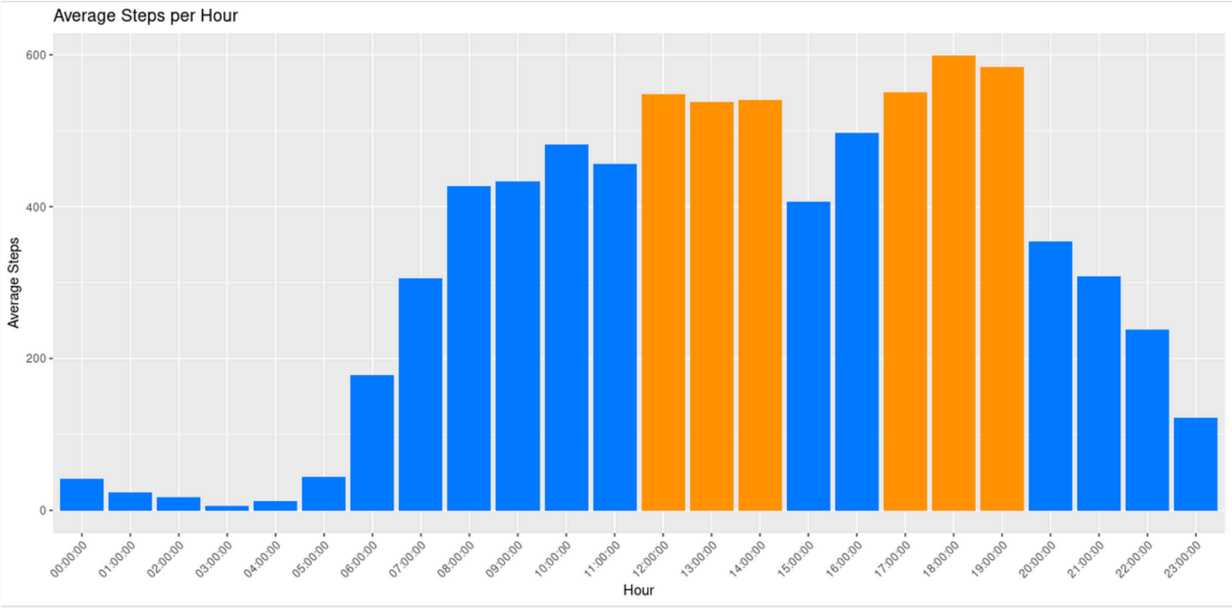

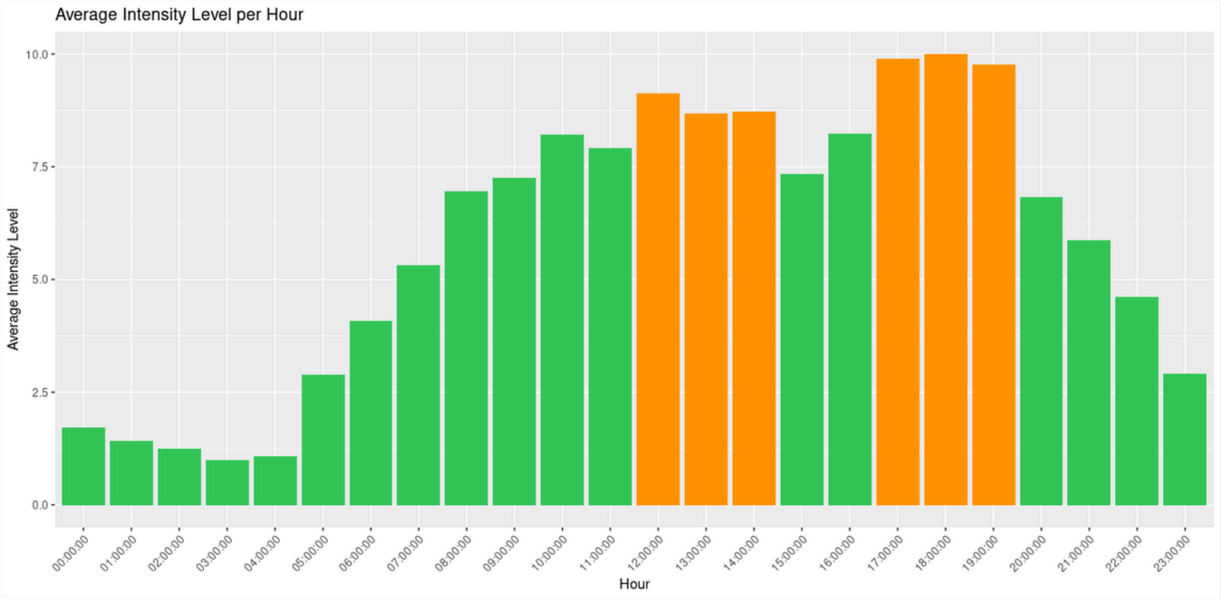

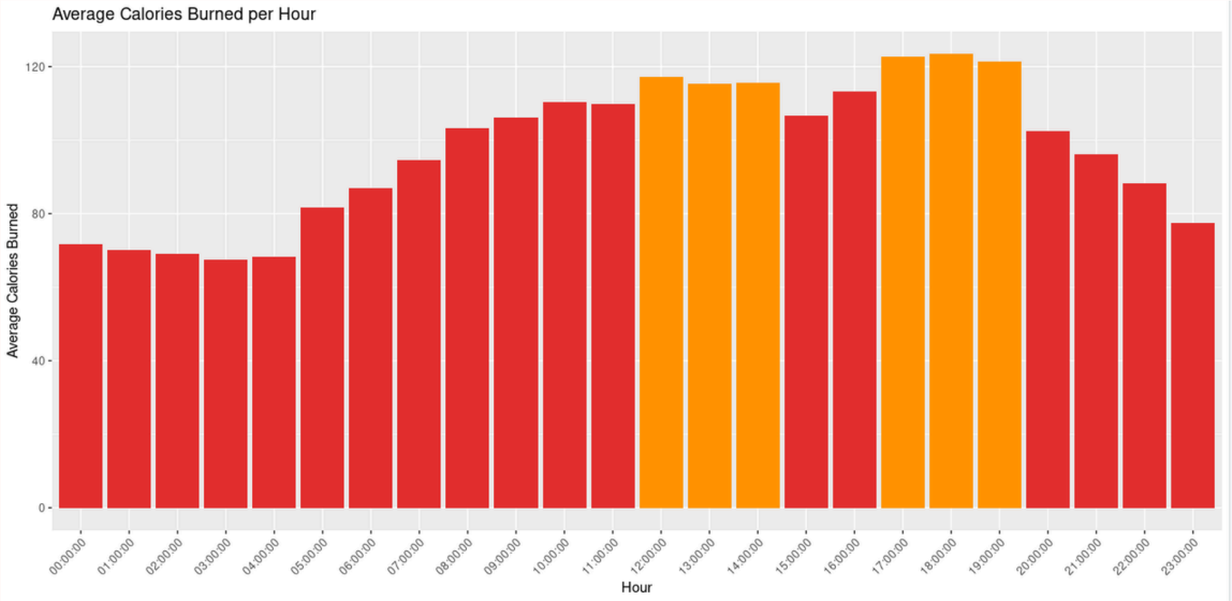

- Highest steps recorded are 12PM to 2PM and 5PM to 7PM
- Highest intensity levels are 12PM to 2PM and 5PM to 7PM
- Highest calories burned are around noon and 5PM to 7PM

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Recomendation

Market segment: People who care about sleep quality

Focus area:

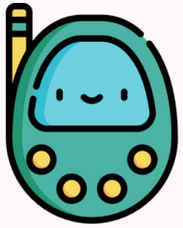
- Sleep quality improvement
- Increase movements

Marketing platform: Social media especially Tiktok

- Tiktok Advertising
- Engage/collab with fitness influencers

Recommendations:

- Short meditation videos/music and light stretching exercise before bed, along with training tips, workout videos
- Add feature to track how users sleep (breakdown of duration, REM cycle)
- Encourage people to get more steps in by keeping streak by having a rewarding mechanism (for example, virtual pet)



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Conclusion

- There are large portions of people who do not sleep enough and exercise enough to the recommended standard
- Participants exhibit minimal physical activity during standard work hours (9 AM – 5 PM)
- Fewer steps lead to fewer sleep hours, which is strongly linked to low quality of life

We can help people by implementing new features that focus on increasing movements throughout the day and improving sleep quality at night.

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